

# Francisco Rau

## Football Data Analyst - Recruitment & Player Valuation

+31 6 38859519 | franciscorau08@gmail.com | franciscorau.com | linkedin.com/in/francisco-rau | Tilburg, NL

### Summary

---

Data analyst with an MSc in Data Science & Society and a BSc in Economics (Tilburg University), building recruitment analytics as my own research programme. Alpha Scout is a player-valuation and recruitment platform I built end to end - 23,445 players across 155 leagues today, ~98,000 player-seasons of history with observed 12-month outcomes, a fair-value model and calibrated probabilities. Live demo via franciscorau.com. Professionally I spent a year turning operational data into production decision tooling. I care most about work that survives its own audit: the numbers below include the ones that got worse when I checked them properly.

### Selected Football Work

---

#### Alpha Scout - Player Valuation Platform (beta)

2024-2026 - Python, statistical validation, Cloudflare

- Built the full pipeline solo: self-healing browser harvesting of per-match ratings and market-value histories, entity resolution across sources, modelling, calibration, product and design. Coverage today: 23,445 players, 155 leagues, weekly automated refresh; ~98,000 player-seasons of history behind the models.
- Hedonic fair-value model prices every player from performance, age, minutes, league difficulty, contract and club context ( $R^2$  0.74 on live data). The gap between fair value and market price is the scouting signal - and it is validated per league, with the backtest published including the leagues where it is weak.
- Calibrated 12-month probabilities per player -  $P(\text{value rises})$ ,  $P(\text{falls})$ ,  $P(\text{rises } 50\%+)$  - from a gradient-boosting classifier with isotonic calibration. Rolling out-of-sample design (train to  $y-2$ , calibrate on  $y-1$ , test on  $y$ ): AUC 0.87-0.90, Brier skill score +0.33 to +0.48, clean reliability curves.
- Every probability ships alongside its twin computed WITHOUT the valuation gap, so a card can state honestly when a call is driven by age and price rather than by the model's signal.
- Forward-tested, not curve-fit: players the model flagged as undervalued were signed 1.7x more often than comparable peers in the following transfer window (19.7% vs 10.7%), measured on picks ranked BEFORE the window opened. An earlier version reported ~3x; auditing it found look-ahead bias in how prices and season vintages were dated, and I rebuilt it. The honest number is narrower - and defensible.
- Tested an additional data provider and dropped it: forward testing showed no predictive gain (IC 0.359 without vs 0.352 with), so it stayed out of the model.
- The model knows its own limits: it declines to price players above EUR 60M - in 98,000 historical player-seasons there is not one comparable case to calibrate on, so it says so instead of guessing. Validity boundaries are stated, not hidden.

### Experience

---

#### Data & Process Analyst - Netvice

May 2025 - Jul 2026 - Tilburg, Netherlands

- Established operational BI as a new, self-owned workstream - a first in the company's ten-year history - owning it from roadmap to daily use on the floor.
- Forecasting model turning planned volume into the exact headcount needed on any future day, 500+ days ahead, with capacity-choke detection and a batch-splitting optimiser.
- Percentile-based pacing metric making individuals comparable across campaigns with very different workloads, combined with outcomes into one performance index - the same normalise-then-rank problem as comparing players across leagues.
- Fully self-directed: each system began as my own proposal - need identified, pitched, built and shipped.

#### Student Researcher - Philips

Sep 2022 - Dec 2022 - Eindhoven, Netherlands

- Built linear models estimating adoption of a subscription service across student segments.

### Skills

---

Modelling: gradient boosting, regression, probability calibration (isotonic), rolling out-of-sample validation,

Brier/AUC/reliability, information coefficients, look-ahead auditing  
Engineering: Python (pandas, scikit-learn), data harvesting & entity resolution across sources, PostgreSQL,  
TypeScript/React front ends, Git, AI/agent dev tooling  
Domain: player valuation, transfer-market dynamics, squad and contract context, league-strength adjustment

## Education

---

|                                                                             |             |
|-----------------------------------------------------------------------------|-------------|
| <b>MSc Data Science &amp; Society (Business Track) - Tilburg University</b> | 2023 - 2024 |
| <b>BSc Economics - Tilburg University</b>                                   | 2020 - 2023 |

## Languages

---

English (native), Spanish (native), Dutch (A2, actively learning)